

SECTION C

Answer **ALL** questions on your question paper (50 marks)

S/N	Question	Answer
1	Choose the most <u>correct</u> statement: (A) The wind shifts clockwise with the passage of a mid-latitude cyclone in the Southern Hemisphere. (B) Southerly meridional flow is associated with sharp drops in temperature over most of South Africa, especially the southern regions. (C) The summer thermal trough over the interior of South Africa is a baroclinic feature. (D) Easterly waves are baroclinic disturbances. (E) Cut-off lows are caused by a deepening of the easterly wave trough	
2	Troughs in the Southern Hemisphere are usually characterised by: (A) positive relative vorticity and negative horizontal divergence (B) negative relative vorticity and negative horizontal divergence (C) positive relative vorticity and positive horizontal divergence (D) negative relative vorticity and positive horizontal divergence	
3	A dynamical atmospheric model is based on equations derived from physical laws. True or False.	
4	Which of the following idealised winds is <u>super-geostrophic</u>? (A) Atmospheric Boundary Layer balanced flow between uncurved isobars (B) Atmospheric Boundary Layer cyclonic gradient wind (C) Free atmosphere cyclonic gradient wind (D) Free atmosphere anti-cyclonic gradient wind	
5	Which naturally occurring climate forcing does stratospheric aerosol injection seek to <u>mimic</u>? (A) The El Nino Southern Oscillation (B) Solar flares (C) Earthquakes (D) Volcanic Eruptions	
6	If the environmental lapse rate is 10K per km and the temperature at the Earth's surface is 300K, then the air temperature at 2000 m above the ground is: (A) 0K (B) 250K (C) 280K (D) 290K (E) 320K	

7	Among stable air columns with sea-level pressure of 1013hPa and the following surface properties, which would likely have the highest geopotential height at the 500hPa-level? (A) Temperature = 30°C, Dew Point Temperature = 13°C (B) Temperature = 29°C, Dew Point Temperature = 17°C (C) Temperature = 35°C, Dew Point Temperature = 17°C (D) Temperature = 15°C, Dew Point Temperature = 12°C	
8	In terms of Earth's energy balance, the most significant impact that deforestation tends to have is to: (A) Increase transpiration (B) Decrease albedo (C) Increase emissivity (D) Reduce insolation (E) Increase albedo	
9	Storm systems tend to decay when: (A) Clouds grow too tall, and water starts to freeze, thus taking up energy (B) They move over water that is too hot (C) There are strong updrafts in the storm (D) They are cut-off from their moisture source	
10	While sitting on Monwabisi Beach overlooking the ocean on a hot summer's day, Person X starts to feel sand being blown into their face. This is likely to be caused by a: (A) Sea breeze (B) South-West Monsoon wind (C) Land breeze (D) Mountain breeze (E) Urban Heat Island Circulation Cell	
11	As rain falls through warm air with low humidity and is evaporated: (A) energy is released (B) the warm air cools (C) air pressure decreases (D) a warm front forms	
12	On average, air in which of the following circulation cells has the highest specific humidity (q)? (A) The Northern Hemisphere Polar Cell (B) The Southern Hemisphere Ferrel Cell (C) The Tropical Hadley Cell (D) The Southern Hemisphere Polar Cell (E) The Northern Hemisphere Ferrel Cell	

13	Surface Southern Hemisphere cyclones are characterised by flow that is: (A) Clockwise, outward and upward (B) Clockwise, inward and downward (C) Clockwise, inward and upward (D) Anti-clockwise, outward and downward (E) Anti-clockwise, inward and upward	
14	Which of the following applies to high pressure systems? (A) They form often along the ITCZ (B) They are associated with moist convection (C) They cause subsidence and divergence at the surface (D) They commonly spawn cold and warm fronts	
15	Which of the following cloud types is a mid-level cloud? (A) Cirrocumulus (B) Nimbostratus (C) Altostratus (D) Stratocumulus	
16	Which of the following is a statement about climate? (A) An intense thunderstorm occurred last night (B) The average temperature in Cape Town over the past 50 years was 290K (C) Yesterday was an extremely hot and humid day in Limpopo (D) Fog is expected to form around Gqeberha today.	
17	Cold fronts are: (A) A tropical weather system (B) A baroclinic weather feature (C) A nearly barotropic weather feature (D) An exclusively Southern Hemisphere phenomenon	
18	A change of state directly from a gas to a solid is called: (A) Sublimation (B) Deposition (C) Condensation (D) Evaporation	
19	When upper-level convergence of air above a surface high pressure area is stronger than the divergence of surface air, the surface pressure will: (A) Increase (B) Stay the same (C) Decrease (D) Become negative	

20	Which of the following is correct regarding wind in the presence of friction in the atmosphere: (A) Friction is strongest in the atmospheric boundary layer, where it slows winds down, but winds blow in the same direction as if there were no friction (B) Friction only acts above the tropopause (C) Friction generally increases with height due to increasing atmospheric density (D) Friction speeds winds up in the free atmosphere (E) Friction has a significant impact on both observed wind speed and direction near the Earth's surface, as it also reduces the Coriolis deflection	
21	For a given pressure gradient J at a latitude Φ, if the Earth completed a full rotation about its axis in 12 instead of 24hrs, the geostrophic wind speed would: (A) show no change (B) double (C) quadruple (D) halve	
22	If an air parcel is given a small push upward and it falls back to its original position, the atmosphere is said to be: (A) stable (B) unstable (C) in geostrophic balance (D) neutrally stable	
23	At 17°C, the atmosphere is saturated with water vapour. If the temperature is increased to 27°C, with no addition of water vapour, one may conclude that the <u>dew point depression</u> is: (A) 17°C (B) 27°C (C) 0°C (D) 44°C (E) 10°C	
24	An anticyclonic gradient wind speed is: (A) Sub-geostrophic (B) Equal to the Geostrophic Wind Speed (C) Super-geostrophic (D) Centripetal	
25	A low pressure on a surface of a given height corresponds to a _____ on a corresponding pressure surface. (A) low pressure (B) low height (C) high pressure (D) high height	

26	A segment of a surface cold front extends along an essentially straight line from 30°S, 10°E to 35°S, 15°E. The thermal wind along the front will be approximately: (A) North-Westerly (B) Westerly (C) South-Westerly (D) South-Easterly (E) North-Easterly	
27	If I run a Global Circulation Model (GCM) at horizontal resolution of 10 km resolution and a Regional Climate Model at horizontal resolution of 150 km, which of the models will be appropriate for simulating 60 km wide sea-breeze (A) GCM (B) RCM (C) Both Models (D) None of the Models	
28	The temperature air must be cooled to in order to become saturated is the (A) Virtual Temperature (B) Dew-Point Temperature (C) Wet-Bulb Temperature (D) Dry-Bulb Temperature	
29	If the absolute temperature of an object triples, the maximum energy emitted changes by a factor of (A) 1/3 (B) 1 (C) 3 (D) 9 (E) 81	
30	The method by which energy is transferred from one body to another with which it is in contact is called: (A) Radiation (B) Convection (C) Conduction (D) Latent Heat	
31	An upper-level pool of cold air that has broken away from the main flow is called a (A) Cut-Off Low. (B) Short-wave trough (C) Long-wave Rossby wave (D) Coastal low	
32	When the air is saturated, which of the following statements is not correct: (A) The air temperature equals the wet-bulb temperature (B) The relative humidity equals 100% (C) A decrease in temperature will cause condensation to occur (D) The virtual temperature is lower than the parcel temperature	

33	Emission inventories are useful for which of the following? (A) Establishing regulations (B) Health risk assessments (C) Establishing monitoring strategies (D) Air quality modelling (E) All of the above	
34	Air pollutant emissions can be from which of the following sources? (A) Natural (B) Anthropogenic (C) Gas (D) Stationary (E) All of the above	
35	What is the largest anthropogenic driver of air pollution? (A) Agricultural activities (B) Industrial activity (C) Domestic burning (D) Energy generation (E) Vehicles	
36	Which of the following meteorological factors can make air pollution worse? (A) Cut off lows (B) Large rain events (C) Temperature inversions (D) Cold winter days	
37	Surface convergence associated with an upper tropospheric jet streak occurs most frequently beneath which two quadrants of the jet streak? (A) Equatorward entrance and exit zones (B) Poleward entrance and exit zones (C) Equatorward entrance and poleward exit zones (D) Poleward entrance and Equatorward exit zones	
38	Vehicle emissions are major primary sources of which of the following pollutants? (A) O ₃ (B) PM ₁₀ (C) CO (D) None of the above	
39	Which of the following pollutants impact negatively on both human health and ecosystems: (i) CO (ii) O₃ (iii) SO₂ (iv) NO₂ (A) i (B) i & iii (C) i, ii, iii (D) ii, iii, iv	

40	If an initially irrotational air column moves westward extending down southern Africa's western escarpment, conservation of absolute vorticity will: (A) Lead to surface divergence (B) Increase the surface air density (C) Change the sign of the Coriolis parameter, f (D) induce anti-cyclonic circulation (E) induce cyclonic circulation	
41	Tropical Temperate Troughs: (A) Occur when mid-latitude waves become linked with low-latitude easterly waves. (B) Result from the conservation of absolute vorticity and topographical forcing. (C) Are upper air features driven by gradient wind flow. (D) Cause cut-off lows.	
42	Choose the correct answer: (A) Cut-off lows are often associated with heavy rainfall (B) Coastal lows are deep cyclonic systems that cause berg winds and fog (C) The easterly wave trough over South Africa is a winter feature associated with surface convergence to the east and divergence to the west (D) Subtropical anticyclones are associated with an elevated inversion which facilitates rainfall in the interior of South Africa (E) Particular synoptic types always cause the same weather at a particular location.	
43	Which of these is not a large-scale climate teleconnection: (A) El Nino Southern Oscillation (B) Indian Ocean Dipole (C) Pacific Decadal Oscillation (D) Tornadoes	
44	The ozone layer is found in the: (A) Troposphere (B) Stratosphere (C) Thermosphere (D) Mesosphere (E) Mesopause	
45	The most common impact of an El Niño on southern Africa is: (A) Hot, dry summers over most of the region (B) Mild, wet summers over the interior (C) Cold, dry winters along the coast (D) Cold, wet springs over the interior	
46	Which one of the following processes releases latent heat? (A) Evaporation (B) Freezing (C) Melting (D) Sublimation	

47	As the warm sector of an eastward propagating surface cold frontal system moves across the cold Benguela current off South Africa's west coast, how will diabatic processes affect the strength of the cross-frontal temperature gradient: (A) strengthen it (B) weaken it (C) have no measurable effect (D) strengthen it at night only	
48	Compared to warm bodies, colder bodies tend to emit radiation that has (A) Higher energy and shorter wavelength (B) Lower energy and shorter wavelength (C) Lower energy and longer wavelength (D) Higher energy and longer wavelength	
49	Choose the correct statement, <i>Parameterization in climate models</i>: (A) is used to increase grid resolution (B) is used for representing processes that are too small to be captured by the modelling (C) is a tool for post-processing data (D) is used as boundary conditions for the model	
50	Which of the following has the lowest albedo? (A) Snow (B) Sand (C) Tar roads (D) Lakes (E) Grass	